

Shawn Ray

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[shray4](#) 
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Skills

Scientific Toolkit: Python - NumPy, SciPy, Matplotlib, NbodyKit, Pandas, Pytorch, Pypelid

Developer Tools: Git, GitHub, VS Code, L^AT_EX, Jupyter Notebook

Data & Software Engineering: SQL, Rust, C/C++, GitHub Actions, pytest, HPC, FITS, CSV, HDF5, DAT

Education

Master's of Science in Astrophysics

2026

City University of New York (CUNY) – Graduate Center | New York, New York

Thesis - *Considering Numerical Relativity for Upgrades in McFACTS Merger Dynamics*

Bachelor's of Science in Physics

2023

Oklahoma State University | Stillwater, Oklahoma

Research Experience

Astrophysics Researcher

2024 – Present

CUNY Graduate Center | New York, New York

- Upgraded the Python based [McFACTS](#) framework, an open-source Monte Carlo software for modeling binary black hole mergers within active galactic nuclei (AGN), by implementing a Numerical Relativity Surrogate model yielding more physically accurate descriptions of post-merger black hole spins, masses, and kick velocities.
- Improved McFACTS algorithms by refactoring the OOP code base, integrating software testing with pytest, and maintaining version control package dependencies with GitHub.
- Managed the standardization and overhaul of McFACTS documentation to meet release standards and facilitate a successful public release.

Astrophysics Researcher

2022 – 2023

Oklahoma State University | Stillwater, Oklahoma

- Analyzed Fermi Gamma-ray Space Telescope Gamma-ray Burst (GRB) data using FITS, HDF5, and CSV files to help create a standard relationship between a GRB's total energy and distance using T90 durations
- Successfully managed concurrent cross-field projects in GRB data analysis, galaxy catalog simulations for interloper identification, and restoration of the university observatory.

Maximizing Student Potential Intern

2023

NASA Jet Propulsion Laboratory | Pasadena, California

- Improved spectral interloper identification in simulated Euclid telescope data catalogs using Pypelid, a Python based simulation pipeline for adding observational effects into simulated catalog data.
- Generated power spectra and galaxy redshift verification tests of simulated Euclid telescope observations using Pypelid and Gattaca, a HPC cluster, to identify the potential observational baryonic acoustic oscillations peaks made from 164 billion simulated galaxies.
- Optimized Pypelid to allow for generalized use with simulated Nancy Roman Space Telescope catalogs.

Electrical Engineering Researcher

2021 – 2022

OSU Department of Electrical Engineering | Stillwater, Oklahoma

- Founder of the first Cube Satellite program in Oklahoma
- Managed the development of CubeSat life cycle processes for both space and ground station operations.
- Designed, built, and maintained an autonomous ground station using C++ for motor control, signal processing, and transmission.

Summer Projects

Dunlap Institute Astronomical Instrumentation Summer School

2023

University of Toronto | Toronto, Ontario, Canada

- Learned the basics of astrophysics instrumentation through hands on lab sessions and lectures, covering major sections of the EM spectrum in the context of astrophysical studies such as radio, infrared, optical, and gamma-rays

Hardware Developer (Co-op)

2022

IBM | Durham, North Carolina

- Developed Python based GUI for more efficient interaction and autonomous use of the C++ based Power Line Disruption (PLD) testing equipment.
- Employed PLD testing methods to determine the effects of RF emissions on the hardware of IBM data servers.

Publications

First Authored

Shawn Ray, Keefe Mitman; Vera Delfavaro; Emily McPike; Barry McKernan and K.E. Saavik Ford (In Prep). “Considering Numerical Relativity for Upgrades in McFACTS Merger Dynamics”. In: *The Astrophysical Journal*.

Co-Authored

Vera Delfavaro; **Shawn Ray**; Harry E. Cook; Kaila Nathaniel; Barry McKernan; K. E. Saavik Ford; Jake Postiglione, Emily McPike and Richard O’Shaughnessy (Aug. 2025). “Prospects for the formation of GW231123 from the AGN channel”. In: *Physical Review Letters, In Review* 2508.13412. URL: doi.org/10.48550/arXiv.2508.13412.

Harrison E. Cook; Barry McKernan; K.E. Saavik Ford ; Vera Delfavaro; Kaila Nathaniel; Jake Postiglione **Shawn Ray**, Emily McPike and Richard O’Shaughnessy (Oct. 2025). “McFACTS II: Mass Ratio–Effective Spin Relationship of Black Hole Mergers in the AGN Channel”. In: *The Astrophysical Journal* 993.2, p. 19. URL: doi.org/10.3847/1538-4357/adfd56.

Vera Delfavaro; K.E. Saavik Ford; Barry McKernan; Harrison E. Cook; Emily McPike; Kaila Nathaniel Jake Postiglione, **Shawn Ray** and Richard O’Shaughnessy (Oct. 2024). “McFACTS III: Compact Binary Mergers from Active Galactic Nucleus Disks over an Entire Synthetic Universe”. In: *The Astrophysical Journal* 989.67, p. 12. URL: doi.org/10.3847/1538-4357/ade4c1.

Barry McKernan; K.E. Saavik Ford; Harrison E. Cook; Vera Delfavaro; Emily McPike; Kaila Nathaniel Jake Postiglione, **Shawn Ray** and Richard O’Shaughnessy (Oct. 2024). “McFACTS I: Testing the LVK AGN channel with Monte Carlo For AGN Channel Testing Simulation (McFACTS)”. In: *The Astrophysical Journal* 990.217, p. 17. URL: doi.org/10.3847/1538-4357/adf114.

Presentations

Invited Talks

Black in Astro - Black Space Week: McFACTS findings	Jun 2026
Columbia Astronomy Pizza Lunch: McFACTS findings	Mar 2026
LISA Consortium Monday Call: McFACTS findings	Dec 2025
CUNY Astrofest: McFACTS findings	Sep 2025
Harvard CfA – Lars Hernquist Group Meeting: McFACTS findings	Aug 2025
AstroOnTap: Upcoming T-Coronae Borealis nova	Sep 2024
AMNH Astro Summer Camp: K-12 talk on the solar system and black holes	Jul 2024
Stanford SURF Scholar Lightning Talks	Aug 2021

Conference Talks

American Astronomical Society (AAS): McFACTS findings	Jan 2026
Dynamix Conference: McFACTS findings	Jun 2025
OK-LSAMP Symposium: T90 GRB findings	Sep 2023
NCUR Research Conference: T90 GRB findings	Mar 2023
NCUR Research Conference 2020 [Cancelled due to COVID-19]	Mar 2020
OK-LSAMP Symposium	Oct 2019

Conference Posters

2026 LISA Symposium: McFACTS findings	Jun 2026
Gravitational Wave Physics and Astronomy Workshop 2025 (GWPAW): McFACTS findings	Dec 2025
OK-LSAMP Symposium	Oct 2020
Oklahoma Research Day	Mar 2020
Emerging Researchers National (ERN) Conference	Feb 2020
Louis Stokes Midwest Regional Center of Excellence (LSMRCE) Annual Conference	Oct 2019
American Indian Science and Engineering Society (AISES) National Conference	Oct 2019
OK-LSAMP Symposium	Oct 2019
McNair Gallery of Engagement	Jul 2019

Awards/Fellowships

Society of Indigenous Physicists Fred Begay Fellowship	2026
CUNY GC BrainE Blitz Lightning Competition: McFACTS findings – 2nd Place	2025
Oklahoma State University Physics Department – Best Undergraduate Researcher	2023
AISES Lighting the Pathway Fellow – Academic honor supporting future Indigenous faculty	2021 – 2023
McNair Research Scholar (\$2.5k/yr) – Academic honor	2019 – 2023
OK-LSAMP Research Scholar (\$2.5k/yr) – Academic honor	2019 – 2023
Cobell Scholar – Academic scholarship supporting Indigenous students (\$5k/yr)	2018 – 2023

Outreach

Secretary	2022 – 2023
Oklahoma State University Chapter of Society of Physics Students (SPS)	
- Managed student events such as star-watching parties and outreach physics demos. - Mentored discussions and preparations for the 2023 International Physics Tournament.	
Vice President	2018 – 2023
Oklahoma State University Chapter of American Indian Science and Engineering Society (AISES)	
Directed general meetings involving guest presenters from outside companies, institutions, and student support services around campus.	
Vice President	2020 – 2023
Society for the Advancement of Chicano's and Native Americans in Science (SACNAS)	
Managed science demo outreach events, general meetings, and organizational recruiting for the Oklahoma State area.	

Hobbies ^_^

Jewelry Designer/Creator, Baking, Cooking, Reading Sci-Fi, Crocheting, Hiking, Painting